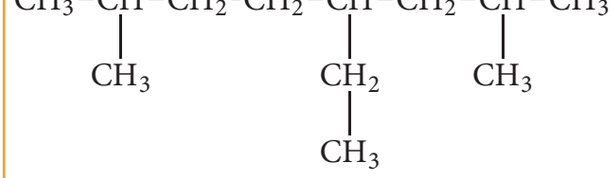
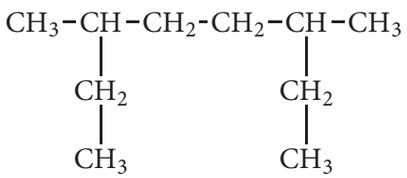
7		4 - Ethyl - 2,7 - Dimethyloctane

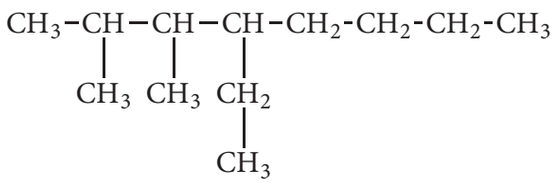
Evaluate Yourself

2) Give the IUPAC name for the following alkane.

a)



b)



How to draw structural formula for given IUPAC name :

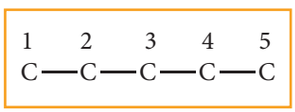
After you learn the rules for naming alkanes, it is relatively easy to reverse the procedure and translate the name of an alkane into a structural formula. The example below show how this is done.

Let us draw the structural formula for

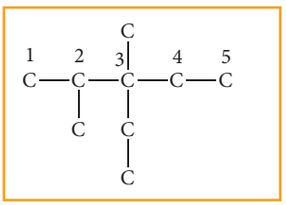
a) 3-ethyl-2,3-dimethyl pentane

Solution:

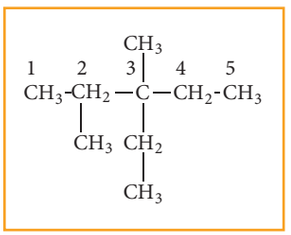
Step: 1 The parent hydrocarbon is pentane. Draw the chain of five carbon atoms and number it.



Step :2 Complete the carbon skeleton by attaching the alkyl group as they are specified in the name. An ethyl group is attached to carbon 3 and two methyl groups are attached to carbon 2 and 3.



Step: 3 Add hydrogen atoms to the carbon skeleton so that each carbon atoms has four bonds



Evaluate Yourself

3) Draw the structural formula for 4,5 -diethyl -3,4,5- trimethyl octane

13.2.1 Preparation of alkanes:

Alkanes are not laboratory curiosities but they are extremely important naturally occurring compounds. Natural gas and petroleum (crude oil) are the most important natural sources. However, it can be prepared by the following methods.

Unit 13 - jagan.indd 183 22-01-2021 12:51:27